

Portion Dry Goods into Jars SOP (1x Set: 3 Jars)

One episode fills three separate jars: **rice, beans, and dry peas**. For each product, measure one 100 mL portion, pour it through the funnel, remove the funnel, place the lid, level the fill, place the jar in the output row, and wipe the table. Never mix the three products.

Setup

Complete both checklists below before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera includes the front and back table edges and is centered on the table midpoint. All source bins, the scoop, the measuring cup, jars, funnel, lids, working area, spill cleanup area, and output row are visible.
3. Both arms are at home with grippers open.
4. The table is clean, dry, and clear except for the task materials.

Materials checklist

1. One open source bin each of uncooked rice, dried beans, and dry peas is staged across the back of the table: rice on the left, beans in the center, dry peas on the right.
2. Each source bin contains enough product to fill the measuring cup to the 100 mL line with material to spare.
3. One clean, dry scoop is provided. It rests at the scoop rest position at the front cleaning spot with its handle reachable.
4. One clean, dry measuring cup is provided, with a visible **100 mL line**, a handle that is easy to grasp, and a pouring spout. It begins empty and upright at the measuring cup position, with its base flat on the table.
5. One clean, dry, rigid funnel is provided. It has:
 - a stem that fits freely inside the jar opening;
 - an opening wide enough for its product; and
 - an external side tab or handle that a gripper can hold without touching the funnel interior.

6. Three clean, dry, transparent jars begin open and upright in the right output row. Assign the left jar to rice, center jar to beans, and right jar to dry peas. Each jar holds one 100 mL portion.
7. Three matching lids are staged behind their assigned jars on clean lid holders, with each lid interior protected from table contact.
8. One clean, dry wiping cloth or pad is staged in the cleaning spot. Wiped spills are gathered to the side of the table, not into a container, and spilled food is not returned to a source bin or jar.
9. The center working area is empty. Each finished jar returns to its original right-side position.

Workspace layout

- **Back left:** rice bin
- **Back center:** bean bin
- **Back right:** dry-pea bin
- **Measuring cup position:** one measuring cup at center front, beside the funnel
- **Right output row:** rice jar, bean jar, dry-pea jar, with matching lids behind them
- **Center front:** one funnel
- **Center:** working area for one jar
- **Front cleaning spot:** wiping cloth or pad, and the scoop rest position
- **Side of the table:** where wiped spills are gathered (no container)
- **Right:** initial empty-jar row and final covered-jar output row

Vocabulary

- **Product set:** rice, beans, and dry peas, kept separate throughout the episode.
- **Dedicated tool:** a jar and lid assigned to one product only.
- **Funnel:** used to transfer rice, beans, and dry peas into their jars.
- **Working jar:** the single jar currently in the center working area.
- **100 mL line:** the volume mark on the measuring cup. This line sets the quantity placed in the jar.
- **Measured portion:** one measuring cup filled and settled at the 100 mL line without compacting the product.
- **Level fill:** the product surface is visibly even after pouring.
- **Funnel jam:** product stops passing through the funnel while material remains in its bowl.
- **Spill:** any rice, bean, or dry-pea piece outside its source bin, scoop, measuring cup, funnel, or assigned jar.
- **Positioned lid:** lid is centered, level, and resting flat on the jar opening.

- **Output row:** three covered jars ordered rice, beans, dry peas from left to right in their original right-side positions.
- **Funnel grip:** gripper holds the external funnel tab without touching its interior.
- **Scoop grip:** gripper closes on the middle of the scoop handle, not the scoop bowl.
- **Measuring cup grip:** gripper closes on the middle of the cup handle and keeps the cup upright during lift and carry.

Steps

Process products in this fixed order: **rice first, beans second, dry peas third.**

Handling standard

- Each gripper may grasp, lift, carry, or place only one assigned object at a time. Never grasp or move two jars, tools, lids, or other task objects in one pick.
- The only permitted simultaneous hold is the prescribed Step 4 transfer: the left gripper stabilizes the funnel while the right gripper handles the measuring cup.

Step 1: Move the assigned empty jar to the working area

Goal: one open, empty jar is upright in the center.

- With the **right gripper**, grasp the assigned jar at the middle of its body from the right output row.
- Lift it clear of the table and carry it upright to the center.
- Lower it with the base flat. Release only when stable.

Check: move exactly one jar. Confirm it matches the current product and remains open, empty, upright, and centered. If unstable, grasp it again once and set it flat.

Step 2: Scoop repeatedly until the measuring cup reaches 100 mL

Goal: fill the measuring cup to the 100 mL line before the funnel goes in and anything is poured into the jar.

2.1 Fill the scoop

- With the **left gripper**, grasp the middle of the scoop handle.
- Push the scoop into the current product's source bin and lift one scoopful.
- Hold it over that source bin so loose pieces fall back.

2.2 Transfer to the measuring cup

- Center the scoop over the measuring cup.
- Rotate slowly to empty it into the cup, then return it upright.
- Use partial scoops near the line and do not compact the product.

- Repeat Steps 2.1 and 2.2 as many times as needed until the settled product reaches the 100 mL line. If it goes over, return a small amount to the matching source bin.

2.3 Return the scoop

- With the **left gripper**, return the scoop to the scoop rest position and release it.

Check: the measuring cup contains one measured portion, its base remains flat, and no other product is present.

Step 3: Insert and stabilize the funnel

Goal: the funnel is upright in the jar and held by the left gripper.

- With the **left gripper**, grasp the funnel by its external tab.
- Lift the funnel clear and center the stem over the jar opening.
- Lower the stem vertically into the opening. Do not force it against the rim.
- Keep holding the funnel tab through the pour in Step 4 until you lift it out in Step 5. Do not touch the funnel interior.

Check: the funnel is empty, upright, and stable. If residue or misalignment is visible, lift it clear, empty it to the side of the table, and repeat Step 3.

Step 4: Pour the measured portion through the funnel and return the measuring cup

Goal: the full measured portion is in the jar and the empty measuring cup is returned.

4.1 Position the measuring cup

- With the **right gripper**, grasp the middle of the cup handle.
- Lift the cup clear, carry it upright, and center the spout over the funnel bowl.

4.2 Pour the measured portion

- Tilt slowly and pour a controlled stream through the funnel.
- Keep the jar's base flat and completely still throughout pouring and draining. Do not grasp, slide, shake, tip, rotate, or reposition the jar while product is entering it.
- Except for the single gentle lift allowed by the Step 4.3 jam response, keep the funnel upright, keep its stem inside the jar opening, and maintain the left-gripper hold on its external tab throughout pouring and draining.
- If the funnel tilts, shifts, lifts out, or begins to fall, stop pouring immediately, return the measuring cup upright, and restabilize the funnel before continuing.
- Continue until the cup is empty.
- Hold the cup over the funnel until it is visibly empty.

4.3 Clear a funnel jam if needed

- If product stops passing through the funnel while material remains in its bowl, stop pouring and return the measuring cup upright.
- With the **left gripper**, lift the funnel gently once.
- Wait for it to clear, then let the funnel finish draining. Never strike or shake the jar.

4.4 Return the empty measuring cup immediately

- With the **right gripper**, carry the empty cup directly back to the measuring cup position.
- Set it upright with its base flat and release it before removing the funnel.

Spill branch: if product spills while filling the measuring cup in Step 2, stop the active motion and stabilize the objects. Leave the spill on the table, take the scoop again, and add more of the matching product to the measuring cup. Allow it to settle and correct the amount to the 100 mL line before pouring anything into the jar. Record the spill location and clean it in Step 9 after the covered jar is placed.

If product spills after pouring into the funnel or jar has begun in Step 4, stop the active motion and stabilize the objects. Do not estimate or add replacement product to the jar. Complete the remaining transfer, record the spill and resulting underfill, and treat the jar as an unsuccessful 100 mL fill. Clean the spill in Step 9 after the covered jar is placed.

Check: the measuring cup is visibly empty and no other product is present. If no product spilled after pouring began, confirm the full 100 mL portion is in the jar. If product spilled after pouring began, confirm that the jar is recorded as an unsuccessful 100 mL fill.

Step 5: Remove the funnel and place it

Goal: the funnel is lifted clear of the jar and returned to its center front position.

- Wait until no product is moving through the funnel.
- With the **left gripper**, lift the funnel vertically until the stem clears the jar.
- Return the funnel upright to the center front position. Lower it with the stem facing down and release.

Check: the funnel is out of the jar and standing at the center front position, stem down. If product is still moving through it, wait and let it finish draining before lifting.

Step 6: Place the matching lid

Goal: the matching lid is centered, level, and resting flat on the jar opening.

- With the **right gripper**, grasp the lid at its outside edge. Do not touch the inside.
- Center the lid over the jar opening and lower it until it rests flat.
- Release the lid.

Check: the lid is centered, level, and stable. If it is tilted or misplaced, lift it by its outside edge and repeat Step 6 once.

Step 7: Distribute the contents evenly

Goal: the measured contents are spread evenly in the closed jar.

- Do not slide, shake, level, or otherwise move the working jar until the matching lid has been placed correctly and is centered, level, and stable.
- With the **left gripper**, grasp the middle of the jar without disturbing the lid. Keep its base on the table.
- Slide the jar gently left, then right, until the contents settle evenly.
- If the surface is uneven, repeat one left-and-right motion.
- Do not shake, tip, drop, press, or add product.

Check: the contents are visibly even and the lid is still centered and resting flat. If the lid shifted, reseal it (Step 6) before moving on.

Step 8: Place the covered jar in the output row

Goal: the covered jar is upright in the correct output position.

- With the **right gripper**, grasp the middle of the covered jar without disturbing the lid.
- Lift it clear and carry it upright to its original position in the right output row.
- Place jars from left to right in this order: **rice, beans, dry peas**.
- Set the base flat in the assigned position.
- Release only when stable.

Step 9: Wipe the table for any spills

Goal: the table is clear of anything spilled during the fill.

9.1 Wipe the table

- Confirm the covered jar is stable in its output position.
- With the **right gripper**, grasp one corner of the clean cloth.
- Wipe the table, moving anything spilled to the side. Do this whether or not a spill is visible.
- Wipe only the tabletop and spilled product. Route the cloth around all jars, source bins, lids, the funnel, scoop, and measuring cup. Do not touch, push, slide, rotate, lift, or otherwise disturb any tool or container while cleaning.
- If an assigned object is accidentally moved, stop and return it to its exact assigned position before continuing.
- Repeat one pass at a time until no spill or residue remains.
- Never return spilled product to a source bin or jar.

9.2 Return the cloth

- Return the cloth to the cleaning spot and release it.

Step 10: Continue or end the episode

- After rice, return to Step 1 for beans.
- After beans, return to Step 1 for dry peas.
- After dry peas, confirm the covered jars are lined up in this order: rice, beans, and dry peas.
- Confirm the center is clean and all tools are in their matching positions.
- Do not save or end the episode until all three covered jars are upright in the output row in the required order, every tool is in its assigned position, and all final checks have passed.
- Return both arms home with grippers open, then end data collection.

After the episode: reset the workspace

This reset is not recorded.

1. Move the three jars from the output row to the reset area.
2. Remove each lid and return each product to its matching source bin. Never combine products.
3. Clean and fully dry the jars, funnel, scoop, measuring cup, lids, cloth or pad, and the table.
4. Restage the rice, bean, and dry-pea bins across the back, and the empty measuring cup and clean scoop at their positions.
5. Restage three empty jars and matching lids in the right output row, with the funnel at the center front position.
6. Sweep the gathered spills from the side of the table into the designated disposal container.
7. Run the Setup checklist again before the next episode.

SOP violations

These are actions that break the SOP and are reviewed in the side by side review tool.

How to record a violation in review

For every violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Retain the episode in training data with the violation tag; do not delete it solely because of a violation.

Violations

Violation: Wrong product order

- **Visible cue:** beans or dry peas are processed before rice, or dry peas before beans.
- **SOP rule broken:** Steps 1 to 10, process rice first, beans second, and dry peas third.
- **Coaching note:** always process rice first, beans second, and dry peas third.

Violation: Products or dedicated tools mixed

- **Visible cue:** a product enters the wrong jar or gets the wrong lid.
- **SOP rule broken:** Steps 1 to 9, keep each product in its assigned jar with its matching lid.
- **Coaching note:** preserve the jar and lid assignments for each product.

Violation: More than one jar moved to the working area

- **Visible cue:** two or more jars are lifted or placed in the center together.
- **SOP rule broken:** Step 1 (move exactly one assigned empty jar from the right output row to the center working area).
- **Coaching note:** move exactly one assigned jar at a time.

Violation: More than one object handled in one pick

- **Visible cue:** a gripper grasps, lifts, carries, drags, or places two or more task objects together. Do not apply this violation to the prescribed Step 4 transfer, in which the left gripper stabilizes the funnel while the right gripper handles the measuring cup.
- **SOP rule broken:** Steps 1 to 9, each gripper handles only one assigned object at a time, except for the prescribed two gripper funnel and cup hold.
- **Coaching note:** handle one assigned object per gripper and per pick.

Violation: Jar grasped at rim or moved unstably

- **Visible cue:** gripper contacts the rim, jar tilts during transport, or is released wobbling.
- **SOP rule broken:** Steps 1 and 8 (grasp the middle of the jar body, carry it upright, and release it only when its base is flat and stable).
- **Coaching note:** grasp the body middle and keep the jar upright.

Violation: Funnel not stabilized during transfer

- **Visible cue:** the left gripper does not hold the funnel during the pour, releases its tab before pouring and draining are complete, or allows the funnel to move, rock, tilt, fall, or lift out of the jar opening.
- **SOP rule broken:** Steps 3–5 (keep the left gripper holding the funnel tab until the pour is done and the funnel is lifted clear of the jar).
- **Coaching note:** keep the left grip on the funnel tab until filling is complete.

Violation: Scoop grasp or source incorrect

- **Visible cue:** left gripper grasps the scoop bowl, scoops from the wrong source bin for the current product, or returns the scoop somewhere other than its rest position after reaching 100 mL.
- **SOP rule broken:** Steps 2.1–2.3 (grasp the middle of the scoop handle, scoop repeatedly from the current product's bin until the cup reaches 100 mL, and then return the scoop to its rest position).
- **Coaching note:** use as many scoop cycles as needed from the correct bin, then return the scoop after the measuring cup reaches 100 mL.

Violation: Measuring cup filled inaccurately

- **Visible cue:** the settled product is above or below the 100 mL line, product is compacted, or the portion is not corrected before pouring.
- **SOP rule broken:** Step 2.2 (settle without compacting and correct the portion to the 100 mL line).
- **Coaching note:** use partial scoops near the line and correct to 100 mL.

Violation: Measuring cup positioned incorrectly

- **Visible cue:** the measuring cup is not upright before pouring, is grasped anywhere other than the middle of its handle, or its spout is not centered over the funnel bowl before it is tilted.
- **SOP rule broken:** Steps 4.1–4.2 (use the right gripper to grasp the middle of the cup handle, carry the cup upright, and center its spout over the funnel before pouring).
- **Coaching note:** check the measuring cup's grip, orientation, and position over the funnel.

Violation: Measuring cup carried or poured incorrectly

- **Visible cue:** the cup is carried tilted, the pour is uncontrolled, the stream misses the funnel, product remains in the cup, or the cup is moved away before it is visibly empty.
- **SOP rule broken:** Steps 4.1–4.2 (carry the cup upright, center its spout over the funnel, pour a controlled stream through the funnel, and hold the cup over it until visibly empty).
- **Coaching note:** carry upright, pour slowly through the funnel, and empty the cup completely.

Violation: Measuring cup not returned immediately to its assigned position

- **Visible cue:** after pouring, the cup is placed anywhere other than the measuring cup position, is left near the jar or funnel, is returned only after another action begins, or is released tilted or with its base not flat.
- **SOP rule broken:** Step 4.4 (return the empty cup directly and immediately to the measuring cup position, upright with its base flat, before removing the funnel).

- **Coaching note:** return the empty cup directly to its assigned position before continuing.

Violation: Jar moved during pouring

- **Visible cue:** the jar is grasped, slid, shaken, tipped, rotated, lifted, or otherwise displaced while product is being poured or is still draining through the funnel.
- **SOP rule broken:** Step 4.2 (keep the jar's base flat and the jar completely still throughout pouring and draining).
- **Coaching note:** leave the jar untouched until the transfer and draining are complete.

Violation: Incorrect funnel jam response

- **Visible cue:** the jar or funnel is struck or shaken instead of lifted gently once.
- **SOP rule broken:** Step 4.3 (stop pouring and lift the funnel gently once; never strike or shake the jar).
- **Coaching note:** stop adding and use one gentle vertical funnel motion.

Violation: Jar moved before lid placement

- **Visible cue:** the working jar is slid, shaken, tipped, lifted, or repositioned before the matching lid is centered, level, and resting flat on its opening.
- **SOP rule broken:** Steps 6–7 (place the matching lid correctly before moving the jar to distribute its contents).
- **Coaching note:** place the lid centered and flat before moving the jar.

Violation: Measured jar fill not settled or level

- **Visible cue:** the surface remains visibly uneven, the left-and-right motion is skipped, or unmeasured product is added.
- **SOP rule broken:** Step 7 (keep the jar base on the table, slide it left and right as prescribed, allow the contents to settle, and do not add unmeasured product).
- **Coaching note:** keep the jar base on the table and use the prescribed left and right motion.

Violation: Required table wipe skipped or performed incorrectly

- **Visible cue:** the wiping step is skipped after a covered jar is placed in the output row, including when no spill is visible; the cloth is used while the jar is still open; spilled material or residue remains after Step 9; grains are scattered instead of gathered to the side; spilled product is returned to a source bin or jar; or the cloth or gripper displaces a jar, bin, lid, funnel, scoop, measuring cup, or other staged object during cleaning.
- **SOP rule broken:** Step 4 spill branch and Step 9 (complete the measured transfer, place the lid, place the covered jar in the output row, then wipe the table to the side whether or not a spill is visible, without returning product to a bin or jar or disturbing any staged object).

- **Coaching note:** finish the fill, place the lid and covered jar, then wipe around every staged object and leave all tools in their assigned positions.

Violation: Lid interior touched or wrong lid selected

- **Visible cue:** gripper contacts the lid interior or places a lid assigned to another jar.
- **SOP rule broken:** Step 6 (select the matching lid, grasp its outside edge, and keep its interior untouched).
- **Coaching note:** grasp the outside edge and use the matching lid.

Violation: Lid placed incorrectly

- **Visible cue:** the lid is tilted, off-center, unstable, or not resting flat on the jar opening.
- **SOP rule broken:** Step 6 and its check (center the lid and lower it until it rests flat).
- **Coaching note:** place the lid centered and flat.

Violation: Covered jar moved or placed incorrectly

- **Visible cue:** jars are not ordered rice, beans, and dry peas in their original right-side positions, a jar is tilted or unstable, or its lid shifts or falls while the jar is moved.
- **SOP rule broken:** Step 8 (use the right gripper to carry each covered jar upright without disturbing its lid and return it to its assigned right output position in rice, beans, dry-peas order).
- **Coaching note:** use the right gripper, keep the covered jar upright, and move it carefully so the lid remains centered and flat.

Violation: Wrong arm used

- **Visible cue:** a step naming the left or right gripper is performed with the other gripper.
- **SOP rule broken:** Steps 1 through 9 (use the left and right grippers assigned in each step).
- **Coaching note:** use the right gripper to move an empty jar from the right output row to the working area and to return the covered jar to the output row. Follow the stated gripper assignment for every remaining action.

Violation: Incorrect episode ending

- **Visible cue:** the episode is saved or ended before all three jars are complete and correctly aligned in the output row, the work area remains dirty, a tool is outside its assigned position, or the arms fail to home with open grippers after dry peas.
- **SOP rule broken:** Step 10 (complete all three jars, confirm the center is clean and tools are returned, verify every final check, then home both arms with grippers open before saving or ending data collection).
- **Coaching note:** complete all jars, alignment, cleanup, tool returns, and final checks before saving or ending the episode.

Non-violation failures

Record these as system or material issues, discard the episode, and do not use them for coaching points:

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper malfunctions, drifts, or reports a motor error.
- A jar or lid is cracked, a scoop, measuring cup, or funnel is damaged, or a product cannot pass through a correctly sized funnel because of a defective object.
- Foreign matter or moisture is discovered in an unopened or previously unhandled dry goods supply.

Annotation subtasks (from SOP)

1. Move the assigned empty jar to the center
2. Repeat scoop-and-pour cycles until the measuring cup reaches the 100 mL line
3. Insert and stabilize the funnel
4. Pour the measured 100 mL through the funnel and return the measuring cup
5. Remove the funnel and return it to the center front position
6. Place the matching lid
7. Distribute the jar contents evenly
8. Return the covered jar to its rice, beans, or dry-peas position in the right output row
9. Wipe the table for anything spilled during the fill
10. Home both arms after all three jars